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M.Tech DEGREE EXAMINATION, MAY 2025

First Semester

21CSC569J - FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE

(For the candidates admitted during the academic year 2024-2025)

Time: 3 Hours **Max. Marks: 100**

(5 × 20 = 100 Marks)

Answer **any 5** Questions

		Marks	BL	CO
1.	(i) Is String a mutable data type? Also explain the various string operations, indexing and slicing in detail with an appropriate example. (10 Marks) (ii) Create a NumPy array of the following product prices: [100, 250, 300, 450, 150]. Perform the following operations. (10 Marks) a. Find the mean price of the products. b. Filter and display the rows where the Category is "Electronics" and the Total_Sales is greater than 10,000. c. Create a bar plot using Matplotlib that shows the Product_Name on the x-axis and the Total_Sales on the y-axis for the top 5 products with the highest total sales. d. Add appropriate labels for the x-axis, y-axis, and the title for the plot. e. Identify and explain which columns in the "sales_data.csv" dataset are examples of categorical data and which ones are numerical data.	20	3	1

Product_ID	Product_Name	Category	Price	Quantity_Sold	Date
P001	Laptop	Electronics	1000	10	2023-01-01
P002	Chair	Furniture	150	50	2023-01-05
P003	Headphones	Electronics	200	25	2023-02-10
P004	Jacket	Clothing	300	20	2023-03-15
P005	Sofa	Furniture	800	5	2023-04-20

2. Demonstrate how to load a dataset into a Pandas DataFrame and clean the data (handle missing values, drop unnecessary columns) and Perform group by operations and calculate summary Statistics, display the first 10 rows, and perform basic data manipulation (sorting, filtering).

Date	Store	Product	Sales	Quantity
2023-01-01	Store A	Widget A	200	10
2023-01-02	Store A	Widget B	150	5
2023-01-03	Store B	Widget A	300	15
2023-01-04	Store B	Widget C		8
2023-01-05	Store A		250	20
2023-01-06	Store C	Widget A	400	20
2023-01-07	Store C	Widget B	100	2
2023-01-08	Store A	Widget C	150	6
2023-01-09	Store B	Widget B		5
2023-01-10	Store C	Widget C	350	10

Demonstrate how to create different types of plots (line plot, bar plot, scatter plot) using Matplotlib. Provide code examples. Explain how to add titles, labels (x and y axis), and customize colors in these plots. Include examples of customization in your

code. 20 3 2

3. Solve the 8-Puzzle Problem Using A* Search Algorithm. A robot is trying to solve an 8 puzzle game where tiles need to be rearranged to match the final state. The blank tile (represented as -) can move up, down, left, or right to swap positions with adjacent tiles. 20 4 3

The initial state of the puzzle is:

2	8	3
1	6	4
7	-	5

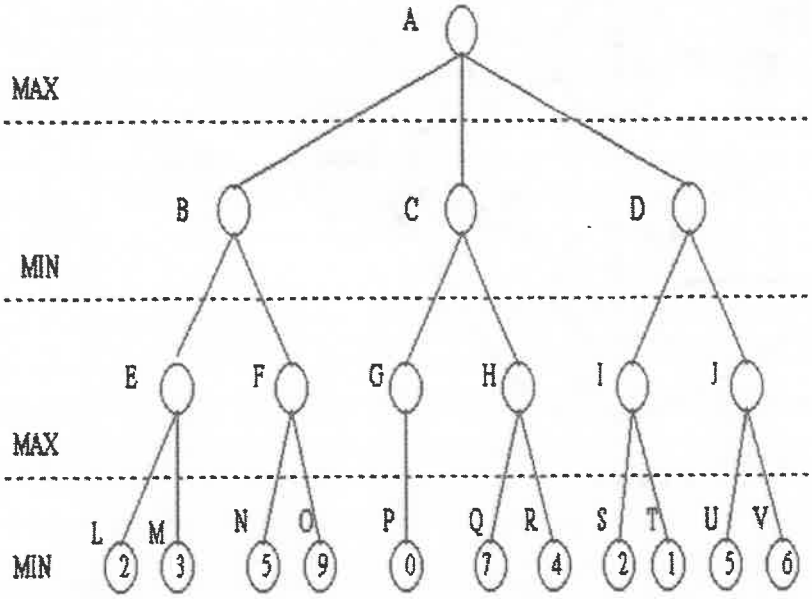
(The blank space - is at position (2,1))
The **final goal state** is:

1	2	3
8	-	4
7	6	5

1. Using A search*, find the optimal path from the initial state to the final state. (10 Marks)
2. What will be the sequence of moves with f(n) calculation taken by the blank tile (-) to reach the final goal state? (10 Marks)

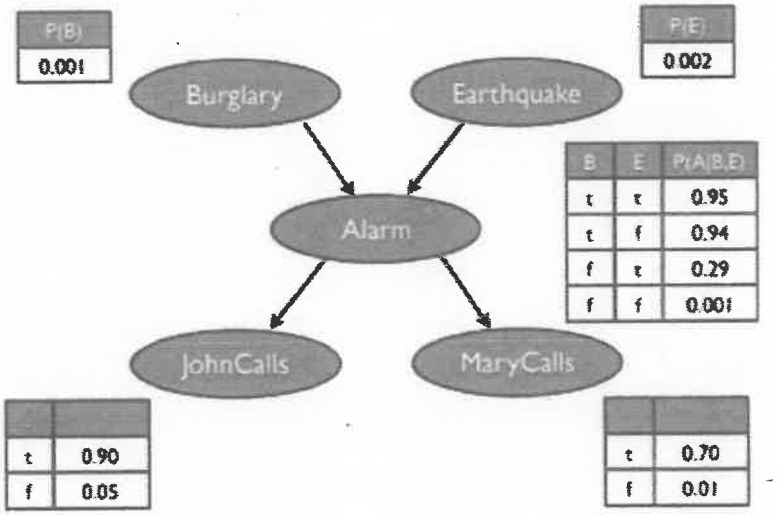
4. (i) Solve the following crypt arithmetic problem and write the step-by-step process. (10M)
Find the value of N+ E + E + D
if: S E N D + M O R E = M O N E Y
(ii) Describe the main idea behind Alpha-Beta Pruning and how it reduces the number of nodes evaluated in a search tree. Find the pruning part of the tree given below((10M)

20 3 4



5. (i) Explain Knowledge based Agents and its Architecture. (6 Marks) 20 3 5
(ii) Brief about Data Mining Techniques in Uncertain knowledge reasoning (7 Marks)
(iii) Write in brief about Bayes theorem in Uncertain knowledge reasoning. (7 Marks)

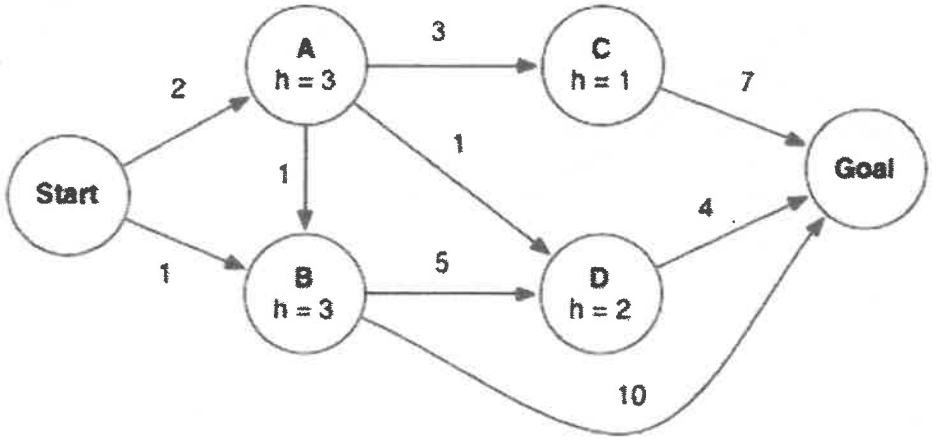
6. (i) Calculate the probability that the Alarm rings (A) given that:(10 Marks) 20 4 6
a. John Calls (J)
b. Mary Calls (M)
c. Earthquake doesn't happen (~E)
d. Burglary doesn't happen (~B)



(ii) Give the syntax, semantic and inference mechanisms in Frames with example (10 Marks)

7. (i) Explain the working principle of Breadth First Search (BFS) and depth first search (DFS) algorithm. Provide a step-by-step illustration with the help of a sample graph. (10 Marks)
(ii) You are developing an AI system to find the shortest path for a delivery robot navigating through a city grid given below with varying terrain costs. Describe how you would implement the A* Search algorithm for this scenario, considering factors such as heuristic functions, node expansion, and path cost evaluation (10 Marks)

20 3 3



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